

Lunar Metropolis: an agent-driven simulation of lunar city growth, and what it says about the balance between surface settlement and subsurface arcologies

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Abstract

We describe *Lunar Metropolis*, an executable model of lunar settlement growth built as a playable simulation, and report what the model says about where people can be housed on the Moon. The model couples an elevation field to a solar-exposure raycast, and from that single coupling derives both the distribution of water ice and the economic value of ground. Three habitat strategies are represented: surface construction, arcologies bored downward into permanently shadowed ice deposits, and arcologies extending along pre-existing lava tubes. We calibrate the model's resource parameters against its own generated worlds, verify its behaviour with a suite of 130 scenarios performing 12,322 simulated days across 265 generated worlds per pass, and compare the three strategies on a common footing. The central result is that subsurface habitation in this model does not save money — it costs between 17 and 76 times more per resident than surface construction — but buys surface area, at up to 2,295 residents per occupied surface tile against 62 for the best surface alternative. A second result is that the three strategies are not equally available: at realistic latitudes, subsurface habitation is effectively a polar privilege, with equatorial mare sites offering a median of 2 ice tiles and no lava tubes at all. We describe the model, its calibration, its verification methodology, and four classes of failure that were detectable only by running it. Everything reported is regenerable from the accompanying software.

1 Introduction

Most models of lunar settlement are written to answer a question that has already been posed. A structural analysis asks whether a given span will stand [1]; a thermal model asks what a given shell costs to keep warm. Such models are precise about a narrow thing.

This work takes the opposite approach. *Lunar Metropolis* is an executable model of a settlement as a whole — terrain, illumination, volatiles, three utility networks, an economy, and a population that responds to all of them — built as a playable simulation and then interrogated as a model. It is deliberately imprecise about every individual mechanism, and in exchange it can be run end to end, tens of thousands of simulated days at a time, across hundreds of independently generated worlds. What such a model is good for is not predicting a number but exposing a *structure*: which constraints bind, which trade against which, and which supposedly free choices turn out to have been made by the terrain generator before the player arrived.

The specific question we put to it here is the balance between building on the lunar surface and building underneath it. The case for going underground is well rehearsed. Burial under regolith

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reduces the galactic cosmic ray dose, though not as simply as one might expect: secondary neutrons, pions and photons produced within the shield itself first *rise* with shell thickness before falling away [4]. An intact lava tube offers a large shielded volume that nobody has to excavate, and the protection is substantial — a Monte Carlo study puts the effective dose equivalent inside a horizontal lunar tube below 1 mSv yr^{-1} , against 416 mSv yr^{-1} on the open surface [6] — in voids whose morphology at Marius Hills was mapped from imagery decades ago [3] and whose presence there is supported by radar sounding [5]. What is less often stated is what the choice *costs*, in a currency that lets it be compared against simply building on the surface instead. That is the question this model is shaped to answer.

Two features of the Moon do most of the work in the model, and both are consequences of illumination. Near the poles the Sun tracks close to the horizon, so modest relief casts enormous shadows, and crater floors that never see the Sun act as cold traps where water ice is stable [7]; the LCROSS impact into Cabeus returned direct evidence of water in such a deposit [2]. The same illumination geometry that puts the ice there also makes that ground worthless to build on: it is dark, cold, and generates no solar power. The model derives both facts from one raycast, and the tension between them is the engine of everything reported below.

We make three contributions. First, we describe the model and the reasoning behind its structure (Sections 2–3). Second, we calibrate its resource parameters against its own generated worlds rather than by hand, and report the calibration (Section 4). Third, we report what the model says about the surface/subsurface balance, and what it says about the automated director that plays it (Sections 6–7). We are explicit throughout about which statements are results of the model and which are assumptions built into it; Section 8 is devoted to the difference.

2 The model

2.1 Scale and structure

The model is a discrete simulation on a 128×128 grid of 16,384 tiles, each carrying an integer elevation on 16 levels, a terrain type, a solar exposure, a dust loading, and optionally a mineral deposit. One tick is one simulated day. The implementation is approximately 12,800 lines of dependency-free JavaScript across 20 modules, governed by 52 named constants collected in a single table so that every tunable quantity is visible in one place.

The choice of a coarse integer grid over a continuous field is deliberate. The model’s purpose is to expose which constraints bind against each other, and that structure survives discretisation; what discretisation buys is the ability to run a complete settlement history in a few seconds, which is what makes the sweeps in Section 6 affordable.

2.2 Illumination, and what follows from it

Elevation is the primary field; everything else is derived. Each tile’s solar exposure is computed by casting rays outward in eight directions and testing whether the local horizon occludes the Sun, with the horizon rise per unit distance — the parameter `sunSlope` — set from the site’s latitude. Near the pole this parameter is small, so even shallow relief occludes, and deep crater floors receive an exposure of exactly zero.

From that one field the model derives, in order:

- **Where ice is.** Ice is placed only where exposure falls below a shadow threshold, with richness increasing as exposure falls further. This reproduces the qualitative structure of polar cold traps [7] without attempting their actual distribution.

- **What power is available.** Photovoltaic output scales directly with local exposure, so the ground that holds ice generates nothing.
- **What ground is worth.** Land value rises with exposure and with elevation, so shadowed floors are also the least valuable ground to build on.

This triple coupling is the model's central design decision. Because all three follow from a single raycast, the antagonism between them is structural rather than imposed: the most valuable resource sits on the least valuable ground, and is hardest to power. No parameter tunes this relationship, and none can switch it off.

2.3 Networks, demand and progression

Three utility networks must independently reach ground before it develops: surface transit tubes, which serve by adjacency; power conduits, which propagate a flood fill from generators and also conduct through developed buildings; and buried atmosphere mains, which coexist with whatever occupies the surface. Population and employment are driven by a three-index demand model in which habitation grows while jobs outrun residents, and commerce and industry grow while residents outrun jobs, with the two employment ratios summing to unity so the system has a fixed point rather than a ceiling.

Colonies advance through four eras gated jointly on peak population and accumulated research. The gate is conjunctive on purpose: population alone would make research optional, and research alone would let a tiny well-funded outpost build a skyline.

3 Three ways to house a colony

The model represents three habitat strategies. They differ in where the volume comes from, and it is that difference — rather than any difference in comfort or technology — that the model is built to price.

3.1 Surface construction

Ground is zoned and the settlement raises buildings on it as demand and land value allow, to a density ceiling set by the current era. A high-density habitation tile at its final stage houses 62 residents. Surface construction is available essentially everywhere: across all site classes tested, the median buildable area lies between about 16,130 and 16,300 of the 16,384 tiles. It is constrained not by where it may go but by land value, by the era, and by the fact that every zoned tile must be within one tile of all three networks.

3.2 Bored arcologies

Four structures excavate downward into permanently shadowed ice. They may only be sited on an ice deposit, which — because of the coupling described in Section 2 — confines them to shadowed crater floors, and they require a level pad of 3×3 or 5×5 tiles.

Each opens as a single gallery and sinks one level at a time, but only while all three networks reach it and the colony is neither short of generating capacity nor of pressurisation. This is the same condition under which zoned ground develops, applied to a structure. A bore that loses its utilities stops where it is and keeps what it has already opened.

Their yield depends on the ice within reach, summed over a radius of three tiles and *shared* with any other bore whose radius overlaps. Two bores placed close together therefore make each

other worse, which makes the distribution of the deposit — not merely the presence of one tile of it — the object of the siting decision.

The four differ in what they return. One is a pure housing structure; one melts its ice and returns a large part of its own pressurisation; one refines and exports, employing more people than it houses and venting dust that degrades the ground downwind; and one, gated behind the final era, is the largest structure in the model.

3.3 Lava tube arcologies

The third strategy takes a void that already exists. Where the terrain generator carves a sinuous rille, the model records the intact tube continuing beneath it: a course of tiles, a width, and a roof depth. A tube arcology is sunk onto that course and then extends *along* it, one tile at a time, under the same utility condition as a bore.

Three properties distinguish it, and together they are the reason it is modelled as its own family rather than as a fifth bore:

1. **The volume is free.** Nothing is excavated, only sealed, lit and pressurised. It is accordingly the cheapest arcology per resident in the model.
2. **The ceiling belongs to the map.** A bore descends to a depth fixed in its own data; a tube arcology stops when it runs out of tube. Across generated polar worlds the usable tube runs from 24 to 81 tiles against a structural reach limit of 64, so on some maps the structure binds and on others the geology does. The player cannot lengthen a tube.
3. **It returns no atmosphere.** It has no ice to crack, so every one of its residents breathes oxygen produced elsewhere in the colony. The cheapest volume in the model is the only one that gives the pressurisation budget nothing back.

Only one class of site in the model generates a rille, and therefore a tube. This is a strong constraint and an intentional one; its consequences are reported in Section 6.

4 Calibration

Two resource parameters were calibrated against the model’s own generated worlds rather than chosen by hand. Both calibrations are reproduced by `experiments/e3-ice-structure.mjs`.

4.1 Ice richness is banded

Sampling every ice tile across eight generated polar worlds ($n = 4,187$) shows that richness is not continuous but falls into narrow bands (Figure 1, left). This follows directly from the derivation: richness is an affine function of solar exposure, and exposure is the output of a raycast over a quantised elevation field, so it takes a small number of discrete values.

The practical consequence is that a richness threshold is only meaningful if it falls between bands. A threshold inside a band selects everything or nothing. This is worth stating because it is invisible from the parameter alone: a richness requirement of 0.62 looks like a mild constraint and is in fact, at polar latitudes where the sampled range is 0.621 to 0.896, no constraint at all on naturally occurring sites. It binds only on ground the player has deliberately levelled out of a partially shadowed slope.

4.2 The yield divisor

A bore’s output scales with a yield derived by dividing the ice within reach by a reference constant and clamping the result to $[0.25, 1.5]$. Choosing that divisor badly is not a matter of balance but of information: too small a value pins sites against the upper clamp, and a saturated site absorbs the penalty for crowding two bores onto one deposit without showing it.

Sampling the raw ice sum at every legal bore site across eight worlds ($n = 380$) gives a median of 9.31 and a maximum of 15.85. Sweeping the divisor (Table 1, Figure 1 right) shows the obvious choice of 8 — which places the median site at unity — pinning 19.5% of all legal sites against the ceiling. The chosen value of 12 is the smallest divisor at which no site saturates, leaving the crowding penalty always visible.

Table 1: Yield divisor sweep over 380 legal bore sites from eight generated polar worlds. The chosen value is the smallest at which no site is pinned against the upper clamp.

divisor	median yield	95th percentile	% at ceiling
6	1.500	1.500	52.6
8	1.163	1.500	19.5
10	0.931	1.379	0.8
12	0.775	1.149	0.0
14	0.665	0.985	0.0
16	0.582	0.862	0.0

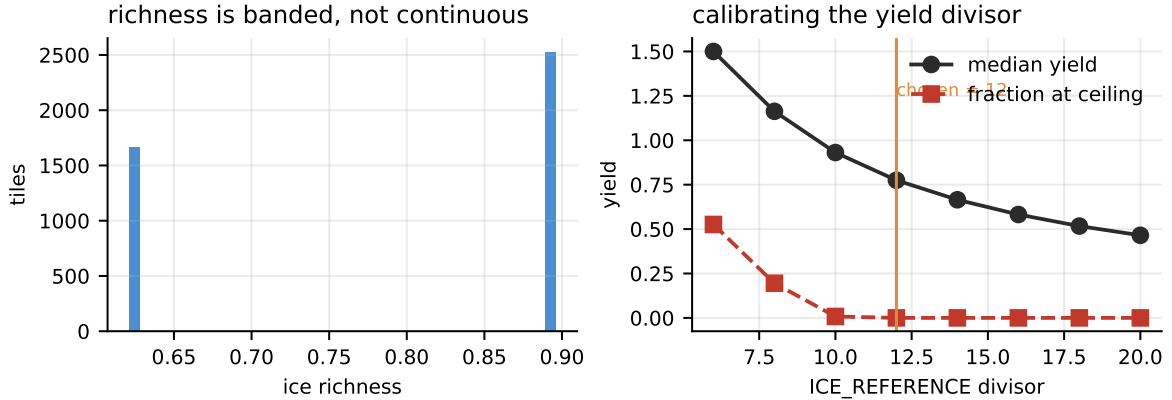


Figure 1: Left: ice richness across eight generated polar worlds, showing the banding that follows from a quantised illumination model. Right: the yield divisor sweep; the chosen value is the smallest at which no site saturates.

5 Verification methodology

5.1 What one verification pass costs

The model is verified by a suite of scenarios, each of which constructs a world, drives the simulation, and asserts on the outcome. At the revision described here the suite comprises **130 scenarios** making **594 assertions**. Executing it performs **12,322 simulated days** — about 34 simulated years — and generates **265 independent worlds**, each a 128×128 field with a full solar raycast.

These figures are counted, not estimated: `e1-model-scale.mjs` wraps the simulation’s tick and world-generation entry points and runs the real suite. We report them because they are the

honest unit of reproducibility for a model of this kind. A statement that the model "passes its tests" means little; a statement that it survives twelve thousand simulated days across two hundred and sixty-five independently generated worlds is a claim a reader can weigh.

5.2 Scenarios as executable claims

Each scenario carries a machine-readable statement of *why* it exists alongside its assertions. This is not documentation but a design constraint: a scenario whose rationale cannot be written down is usually asserting an implementation detail rather than a property of the model, and several were deleted on those grounds during development.

5.3 On counting simulation runs

A reader may reasonably ask how many times the simulation was run in the course of building the model. We can answer that in three parts, and only two of them honestly.

1. **The cost of one pass** is exact and reproducible, and is given above.
2. **Runs from the point of instrumentation onward** are recorded. Every experiment appends to `results/ledger.json` a record of what it executed — simulated days, worlds generated, and the source revision — so the cost of reproducing any figure in this paper is itself a committed artefact.
3. **Runs before instrumentation are not recoverable.** The model was developed over roughly six weeks and 31 commits touching its directory, during which the suite was executed many times, but no count was kept and we decline to estimate one. Reconstructing it from commit timestamps would produce a number with the shape of evidence and none of the substance.

We state this plainly because provenance is exactly the kind of claim that invites a plausible fabrication, and because the fix — instrument first — is cheap and was applied here only in retrospect. The ledger closes the gap prospectively, not retrospectively.

6 Results

6.1 Subsurface habitation is a polar privilege

Generating eight worlds at each of three site classes, at latitudes that would actually produce them and with the solar geometry the model derives from that latitude, gives the availability in Table 2.

Table 2: Habitat availability by site class, over eight seeds each. Values are medians, with the observed range in brackets where it is wide; a median over an even sample averages the middle pair, hence the half values. Buildable surface is essentially unconstrained everywhere; subsurface habitation is not.

site class	latitude	sun slope	buildable	ice tiles	bore sites	tube tiles
polar	−85°	0.201	16,167	527	44.5 [20–86]	60 [24–81]
highland	−50°	0.489	16,134	383	1.5 [0–9]	0
mare	−5°	0.859	16,297.5	2	0 [0–0]	0

The result is stark, and it was not designed in: it falls out of the illumination coupling. At an equatorial mare site the shadow budget is so small that the model places a median of two ice tiles on the entire map, no site qualifies for a bore, and no rille — and therefore no tube — is generated at all. Such a colony has exactly one way to house anyone: build on the surface.

The highland case is subtler and more interesting. Highland sites carry substantial ice — a median of 383 tiles, only a quarter fewer than polar — and yet a median of just 1.5 legal bore sites, with two of the eight seeds offering none at all and the best offering nine. The ice is there; it lies on slopes, and a bore needs a level pad. Resource abundance and resource *accessibility* come apart, which is the sort of distinction a whole-settlement model surfaces and a component model does not.

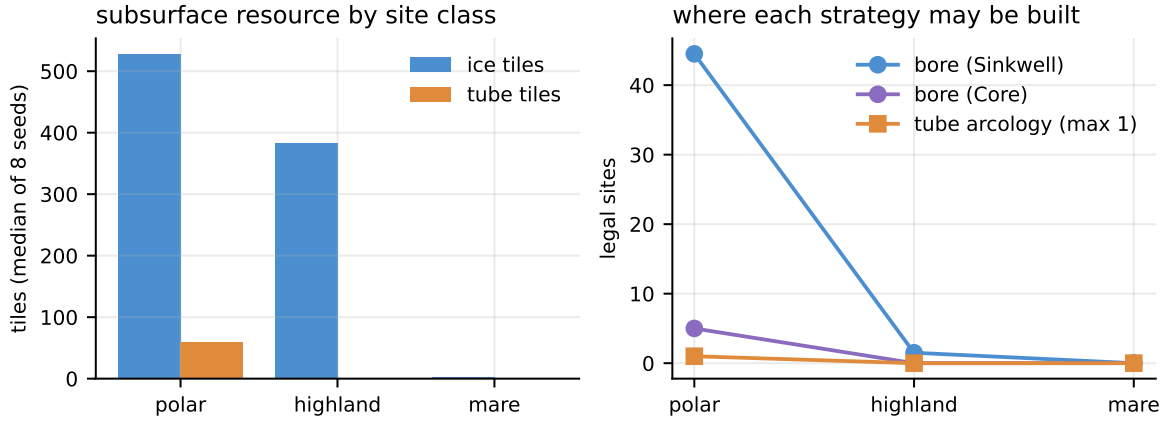


Figure 2: Subsurface resource and legal sites by class. Ice is present at highland sites but largely unusable; tubes exist only at polar sites.

6.2 Arcologies buy land, not money

Costing the three strategies on a common basis (Table 3, Figure 3) gives the paper’s central result, and it inverts the intuition that drove the model’s design.

Table 3: Habitat economics. Surface figures are for one high-density habitation tile at its final stage. Tube figures are the median over eight generated polar worlds; its range is 861 to 2,520 residents depending on the tube the map produced.

strategy	residents	credits/resident	residents/kW	residents/surface tile
surface tile	62	2.4	31.0	62
Sinkwell (bored)	1,100	58.2	32.4	122
Cistern (bored)	480	170.8	12.0	53
Foundry (bored)	405	182.7	7.8	45
Core (bored)	2,450	100.0	25.5	98
Tube arcology	2,295	41.8	52.2	2,295

Surface construction is between 17 and 76 times cheaper per resident than any arcology. Whatever the case for going underground is in this model, it is not capital efficiency.

What arcologies buy is *surface area*. A tube arcology houses 2,295 residents while occupying a single surface tile, against 62 for the best the surface can do — a factor of 37. Bored arcologies achieve 45 to 122 residents per occupied tile once their required pads are counted against them. Since the binding constraint on a surface settlement is the ground within reach of three utility networks, this is the currency that matters, and it is the one in which subsurface habitation wins decisively.

The lava tube is the best deal available on every axis — cheapest arcology per resident, most efficient per kilowatt, and by far the most compact — with two compensating constraints. You

get one, and its size is set by the map: the same structure returns 861 residents on a poor tube and 2,520 on a generous one, a spread of $2.9\times$ that no decision by the player can influence. And it returns no atmosphere, so its residents draw on pressurisation the colony must produce elsewhere.

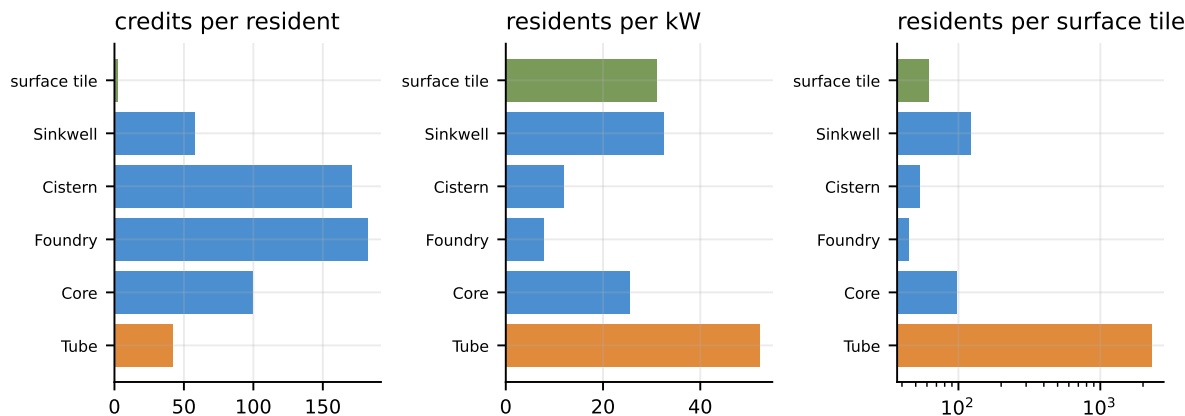


Figure 3: The three strategies on a common basis. Surface construction wins decisively on cost per resident; subsurface habitation wins by a larger margin on residents per surface tile (note the logarithmic axis).

7 Failure modes visible only under simulation

The model includes an automated director that plays the settlement without a human: it lays networks, zones to demand, manages a budget and builds structures. It is useful here as an instrument, because it exercises the model in ways a scripted test does not — and because its failures were failures of the *model’s* reasoning, not of its code.

Four such failures were found by running complete settlement histories. All four passed every placement test throughout, because placement was never what was broken.

7.1 Building what cannot be used

The director initially treated a bore like any other terrain-gated structure: site it wherever the terrain qualifies, the moment it is affordable. But qualifying ground for a bore is, by construction, permanently shadowed crater floor — precisely the ground the settlement has no other reason to expand toward. The director bought structures that its utility networks never reached, and each sat at one level indefinitely: a completed purchase from the outside, and worth nothing.

Over 900 simulated days on three seeds, the naive director built twelve subsurface structures of which **one** ever grew (Figure 4).

7.2 Refusing is not the same as solving

Requiring the utilities to reach a site before building eliminated the waste entirely — no stalled structures on any seed — but at a cost: the director now built only eight structures, because it would not go to the ice at all. The lattice grows toward valuable ground, and shadowed floors are the least valuable ground in the model, so it never arrived.

Only when the director was given an explicit reason to walk a corridor out to the resource did both problems resolve: fourteen structures, all growing, none stalled. The general lesson is that a resource which is valuable only *after* you have reached it will not be reached by an agent that

expands by local value alone. That is a statement about settlement dynamics, not about this implementation.

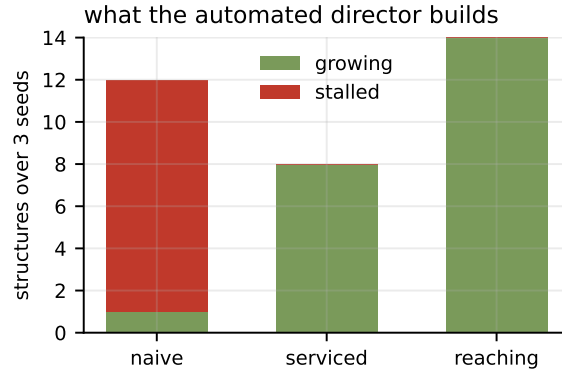


Figure 4: Subsurface structures built by three revisions of the automated director over 900 simulated days on each of three seeds. The naive director builds the most that never work.

7.3 An adjacency predicate used for the wrong question

The model asks "is this ground served by the power network?" using a predicate that counts adjacency — correct for deciding whether a tile may develop. The corridor-building routine reused it to choose where to connect *from*, which is a different question. A transit tube adjacent to a conduit reports itself served and conducts nothing, so corridors anchored on one dead-ended a single tile short, delivering transit and atmosphere to a structure with no power.

The failure is instructive because every tile involved reported itself correctly. Nothing was broken; a predicate was asked a question it does not answer.

7.4 A corridor that blocked its own successor

Corridors laid a transit tube alongside the route for legibility. Excluding the current route from that placement prevented a corridor from blocking itself, but not from blocking a *later* one: on one seed a site became ringed by tubes left by its own abandoned attempts, could never be powered, and the director re-laid the same two-tile corridor to it every 120 simulated days for the remainder of the run.

This one is worth reporting for a methodological reason. On most of the seeds swept during development the same defective revision built its full complement of structures and looked entirely healthy; the fault appeared only on maps where a corridor had to cross ground already occupied by an earlier attempt. That is an argument for sweeping seeds rather than fixing one, and for reporting a spread rather than a best case.

We flag explicitly that this observation, unlike the comparison in Figure 4, is *not* among the recorded experiments. The defective revision existed only in a working tree and was never committed, so `e5-director-ab.mjs` — which extracts its baselines by commit hash — cannot reproduce it. We report it because the failure mode is instructive, and we mark it as unrecorded because it would otherwise carry an authority the other results in this section have earned and it has not.

7.5 A complexity failure hiding behind a logic failure

Finally, the site-selection routine called a placement predicate that itself scans the whole map, once per candidate tile — quadratic in the map area. This was harmless while the director placed

a structure on the first tick it could and then stopped looking. Once it was correctly refusing unreachable sites, it never stopped looking, and the verification suite stopped completing at all. The fix was to order the cheap network test before the expensive placement test, reducing candidates from 16,384 tiles to a few hundred.

A logic change turned a latent complexity bug into a fatal one without either being visible in isolation. We report it because the interaction, not either defect, is the interesting object.

8 Limitations

This is a model of a settlement’s *structure*, not a prediction of any real one. The following limitations are material to how its results should be read.

Parameters are chosen for legibility, not measured. Costs, capacities, growth rates and power draws were selected so that the trade-offs between strategies are visible over a few hundred simulated days. They are not derived from engineering estimates, and the absolute numbers in Table 3 carry no physical meaning. What is meaningful is the *ordering* and the approximate ratios between strategies, since those follow from the model’s structure rather than from any single constant.

The physical grounding is qualitative. The model reproduces the qualitative structure of polar cold traps — ice where illumination never reaches, richer where it reaches least — consistent with observations of the lunar south polar region [2, 7]. It does not attempt the actual distribution, and its ice is far more abundant and more regular than any measurement supports. Likewise, the case for subsurface habitation rests on regolith shielding [4, 6] and on the existence and stability of intact tubes [1, 5], but the model represents neither radiation dose nor structural stability at all. A tube in this model cannot collapse, and its roof depth is recorded but unused. Any statement here about the desirability of going underground is a statement about *land economics under the model’s assumptions*, not about safety.

Illumination is static in time, but not in terrain. Solar exposure is an eight-direction average computed at generation and never recomputed as the Sun moves; only the shading direction is animated. This is defensible at polar latitudes, where the point is precisely that exposure barely changes over a cycle, and progressively less defensible toward the equator — which is where the model’s finding that mare sites cannot support subsurface habitation is least trustworthy.

Exposure *is* recomputed locally whenever the terrain changes, whether by the player sculpting or by a meteor strike. Deposits, however, are seeded once at generation and never re-seeded. The asymmetry is deliberate but has a consequence worth stating: a player can manufacture shadow and cannot manufacture ice, so excavating a crater produces ground that looks like an ice site by every visible criterion and holds none. Whether that is the right model of a cold trap is arguable — ice accumulates in shadow over geological time, not over a season — but it is an assumption, not a derivation.

One tube per world. The model generates at most one lava tube, on one class of site, along the path of a single rille. Real tube systems are networks. The strong result in Section 6 that tubes are polar-only is therefore a property of this generator, not an inference about the Moon; what the model supports is the narrower claim that a habitat strategy whose ceiling is set by geology behaves differently from one whose ceiling is set by budget.

Provenance before instrumentation is lost. As set out in Section 5, the number of simulation runs performed during development was not recorded until late, and we decline to reconstruct it.

The agent is one agent. The director results in Section 7 describe the behaviour of a single heuristic policy. They are evidence about what that policy does, and only suggestive about settlement dynamics in general.

9 Availability, reproducibility and FAIR compliance

9.1 Software

The complete model, the verification suite, the experiments behind every figure and table in this paper, and the source of the paper itself are in one repository under the MIT licence: <https://github.com/dr-richard-barker/LunarSims>. The model has no build step and no runtime dependencies; it is static HTML, CSS and JavaScript, and runs by opening a file.

9.2 Regenerating the results

Every number quoted in this paper is produced by a script in `paper/experiments/`, which writes a JSON record to `paper/results/` and appends to a run ledger. Figures are generated from those JSON records alone — never from the model directly — so a figure cannot silently disagree with the value quoted beside it. The whole chain rebuilds with:

```
make -C paper
```

which runs the experiments, regenerates the figures, and typesets this document. Results are committed, so the paper builds without re-running anything; deleting `paper/results/` forces a full regeneration.

Experiments that compare revisions of the model extract their baselines from version control by commit hash rather than from working copies, so an A/B comparison is reproducible from the repository alone.

9.3 FAIR assessment

Findable The repository carries `CITATION.cff`, `codemeta.json` and `.zenodo.json` at its root, in the locations GitHub and Zenodo read. A DOI is minted on deposit.

Accessible Public repository, MIT licence, no credentials, no dependencies, no build step. The archived snapshot is self-contained.

Interoperable Results are plain JSON with documented fields; the model is standard JavaScript; the manuscript is L^AT_EX with a BibTeX bibliography. Nothing requires a proprietary format or tool.

Reusable The licence permits reuse and modification. Provenance is recorded per experiment (source revision and simulation cost). The limitations in Section 8 state what the model does and does not support, which is a precondition for reuse that is not misuse.

9.4 Fields requiring completion before deposit

The following metadata cannot be derived from the repository and are left as explicit placeholders rather than filled with plausible values. `PREDEPOSIT.md` lists them with the exact file and field

to edit: author ORCID; author affiliation; the release version and tag to archive; and the DOI, which Zenodo assigns on deposit.

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